

[TODO: title] — Near Real-Time Markerless Scoring of the Standardized Drinking Task

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Abstract—[TODO: Write last, ~200 words. Structure: (1) OMC plus biomechanical modeling is the reference standard for upper-limb movement quality but is lab-bound and slow; (2) markerless pipelines have closed much of the accuracy gap, but the differentiable-biomechanics approach needs days of GPU optimization per cohort, so results never reach the clinician during the visit; (3) we present a markerless pipeline — multi-camera 2D pose, robust bundle adjustment, temporal smoothing, geometric measure extraction — that scores the standardized drinking task within X of the end of recording on a single consumer GPU; (4) validated on ten participants and 697 trials against synchronized optical mocap, we reach $r_s \geq 0.86$ on six of eleven Murphy measures and $r \geq 0.96$ on elbow-extension trajectories, without inverse kinematics; (5) accuracy is comparable to offline markerless work while running $X \times$ faster, making same-session feedback feasible.]

Index Terms—markerless motion capture, optical motion capture, drinking task, movement quality, stroke rehabilitation, real-time

I. INTRODUCTION

[TODO: P1 — clinical need: stroke prevalence, upper-limb impairment, the case for frequent objective assessment of movement quality.]

[TODO: P2 — the standardized drinking task and the Murphy measures as the established instrument [1]; OMC plus biomechanical modeling as the reference standard.]

[TODO: P3 — markerless MMC now approaches OMC accuracy [2], removing markers and specialist labor. State the remaining obstacle plainly: latency. Give the concrete figure — the end-to-end differentiable pipeline of [2] required roughly 12 days of GPU processing for 1160 trials, dominated by 2D keypoint extraction over 5800 videos, plus 30 CPU hours of OpenSim IK on the OMC side. A result that arrives days later cannot inform the session that produced it.]

[TODO: P4 — our position. Same task, same measures, same validation protocol, but every stage chosen so per-trial compute is bounded and overlaps with recording. Note the deliberate trade: we compute measures geometrically from 3D keypoints rather than fitting a biomechanical model, which costs some trajectory agreement and buys the latency.]

Contributions:

- A markerless pipeline that produces the Murphy drinking-task movement-quality measures in near real time on a single consumer GPU, with [TODO: a measured per-stage latency budget].
- A validation against synchronized optical mocap on ten participants and 697 trials, reporting per-trajectory and per-measure agreement (Tables I–III).
- [TODO: An account of which measures survive the no- IK , geometric construction and which do not — and why the failures are anatomical (occluded seated hips) rather than algorithmic.]

II. RELATED WORK

A. Movement quality assessment in the drinking task

[TODO: Murphy et al. measures, clinical use, what each is meant to capture.]

B. Markerless motion capture for clinical kinematics

[TODO: Multi-view pose estimation, triangulation, biomechanical fitting. Position [2] as the closest comparison: same task, same measures, OMC-validated — but offline, and IK-based.]

C. Latency in clinical measurement pipelines

[TODO: Little prior work reports latency at all. Argue that time-to-result is a first-class metric for clinical adoption, and that reporting accuracy alone has hidden this axis.]

III. METHODS

A. Dataset

We use the DELTA dataset of [2] unchanged — 40 repetitions of the standardized drinking task [1] per arm, recorded simultaneously by optical mocap at 100 Hz and by five synchronized webcams at 1080p and 60 Hz — and report here only the selection applied on top of it. That selection is stricter about camera quality than an IK-based method needs to be: with no model prior to absorb it, a miscalibrated view propagates straight into the 3D point.

Participants. Ten participants from the same pool as [2]. One was recorded in two sessions under different calibrations, which we treat separately, giving eleven recording units. Recruitment, inclusion criteria and ethics approval are as reported in [2].

Cameras. Cameras were excluded in two passes: first for mis-cut clips, detected by normalized cross-correlation of each cut clip within its uncut session recording, and then for mis-calibration, decided by leave-one-out per-joint reprojection. The order is necessary, as a mis-cut clip produces a smooth, position-dependent reprojection error indistinguishable from a moved camera — both land at 25–30 px. Five cameras were dropped as mis-cut and seven as miscalibrated, leaving three of the eleven units on all five cameras and the rest on three or four.

Trials. 837 clips pass an integrity check that the video, its detections and the resampled mocap span the same window. A trial is then admitted by a temporal gate — cross-correlation ≥ 0.7 , best of {wrist, elbow, shoulder} $\times$ {speed, displacement} and the cup, since raw wrist speed alone rejects sound trials — and a wrist-validity gate of $\geq 50\%$ of frames, together admitting 826 (99%). Of these, 697 carry a ground-truth row with phase windows and enter the measure comparison (per-measure n in Table III), and 687 the trajectory comparison. The remainder are absent from the reference processing rather than failed by ours: the ceiling here is ground-truth coverage.

B. Pipeline

The pipeline runs in six stages.

- 1) *2D body pose.* Stock `yolo26s-pose`, run per frame on every camera.
- 2) *Cup localization.* Stock `yolo26x-seg` locates the cup *once*, at the first frame whose detections agree across at least three cameras; a transformer visual-object tracker (UETTrack-B) then follows it through the remaining frames in each camera. It is never re-anchored: re-seeding from a shared consensus couples the cameras, and can cascade all of them onto the same wrong object, destroying the independence that multi-view voting relies on. The cup trajectory is used *only* to segment the movement into phases (Sec. III-C); no movement-quality measure is computed from it.
- 3) *3D reconstruction.* A multi-view DLT initialises a bundle adjustment that minimises geometric reprojection error over all nine upper-body joints, weighted by detector confidence and robustified by a Huber kernel. The DLT minimises an algebraic residual, which carries no statistical interpretation and exists because it is closed-form; minimising reprojection error is the maximum-likelihood estimate of the joint under Gaussian image noise, and the Huber kernel exchanges that for a heavy-tailed model, so a single grossly wrong detection cannot dominate the fit [3]. Cameras reprojecting beyond 30 px are excluded and the joint refitted on the rest; where too few remain, it is triangulated from all cameras rather than dropped, so a degraded view costs accuracy on

a frame instead of removing the trial. No bone-length prior is used: it holds bone lengths constant but slides joints along the unobservable depth ray, so reprojection error falls while agreement with OMC worsens. [TODO: BLOCKED, needs the compute box: `log n_fallback / n_guarded from refine_trial_ba` over all 826 trials. The BA solve occasionally diverges and currently reverts that joint-frame to the DLT point; the only rate ever measured ($\sim 0.05\%$ of frames, 1 trial in 328) predates the cohort expansion and the miscalibrated participants that cause divergence. Decide then whether to keep the revert or exclude the trial and report the rate, as in [2].]

- 4) *Temporal smoothing.* SmoothNet, applied to the 3D joint trajectories. Its ± 16 -frame window is non-causal, which is what places this stage after the trial rather than alongside it (Sec. III-E).
- 5) *Phase segmentation.* From the cup trajectory; see Sec. III-C.
- 6) *Measure extraction.* Geometric, from 3D body keypoints only; see Sec. III-D.

C. Phase Segmentation

The cup is used here and nowhere else. Segmentation reduces to a single primitive applied to two distance channels: a distance closes, holds flat, then opens again. On the wrist-to-cup distance that pattern is grasp and release — the hand arrives, the distance is flat while the cup is carried, and opens when it is set down. On the cup-to-mouth distance it is the drinking phase, as the cup rises, is held, and is lowered. A plateau describes the relation between two points rather than the motion of either, so the drink boundaries survive the participant moving the head to the cup instead of the cup to the head. Reach onset and the final settle come from the hand leaving and regaining its rest position.

Thresholds are relative: a closing or opening run counts only if it traverses at least 30% of that channel’s own range within the trial, so nothing depends on the size of the workspace or on an absolute distance to the mouth. Boundaries are found in one forward pass, each taken as the first qualifying run after the previous boundary, which makes the phases ordered and non-overlapping by construction.

The cup is reconstructed under a stricter rule than the body. Frames with fewer than three agreeing cameras are discarded rather than forced, because a cup outlier is a different object — another vessel in the room — rather than a displaced joint, and averages to nothing useful. Interior gaps shorter than 0.75 s are filled by a Kalman smoother, never extrapolating beyond the first or last observation; what remains is filled from the wrist, which carries the cup, as a constant offset estimated over the frames where the cup is held. That proxy is applied to the cup-to-mouth channel alone: it makes the wrist-to-cup distance constant by construction, and grasp and release are read from exactly that channel.

Two roles for the cup must be kept apart. The deployed pipeline segments from the cup, but the measure comparison in

Sec. IV gives both systems the same optical phase windows, so that those numbers isolate pose error from phase-classification error. Segmentation is therefore validated separately, as boundary error against the optical phases (Sec. IV).

D. Movement Quality Measures

Measures are computed geometrically from the 3D keypoints; no biomechanical model is fitted and no inverse kinematics is solved. Throughout, position is smoothed and then differentiated, never the reverse, since filtering a raw derivative smears a spike without removing its influence. One threshold is deliberately not inherited from marker-based practice: movement units are counted against a 60 mm/s amplitude rather than the usual 20 mm/s, because the jitter floor of video sits at 30–50 mm/s and the lower value manufactures units. The remaining constants are in the released code.¹

The eleven measures are those of the drinking-task protocol [1]; only the shoulder angles need defining here, since they are computed geometrically rather than read off a fitted model. An orthonormal body frame is built from four torso points — both shoulders and both hips — giving a down axis from the shoulder midpoint to the hip midpoint, a frontal-plane normal, and a lateral axis. Shoulder flexion is the angle of the upper arm from the down axis within the sagittal plane, abduction the angle within the frontal plane. [TODO: add the Alt Murphy planar-definition reference (Table 2) to refs.bib and cite it here rather than leaning on [1].]

Three further departures from [2] are forced rather than chosen. Their trunk displacement follows a sternum marker, a landmark the COCO keypoint set does not contain, so ours is the excursion of the shoulder midpoint from its resting position — and a three-dimensional excursion rather than an anatomical forward projection, because that projection needs a forward normal defined by the hips, which are occluded in a seated participant. The same occlusion is why the body frame is frozen per trial rather than recomputed per frame: an axis that jitters dominates the quantity it is meant to reference. Interjoint coordination changes indirectly, being a correlation involving shoulder flexion. Elbow extension is unchanged.

Peak velocity is the one departure from the smoothing rule above: it differentiates the pose without the additional position low-pass and low-passes the resulting speed instead, because smoothing the position first flattens the peak the measure exists to report.

Drinking-phase shoulder flexion is not reported, no ground-truth column existing for it, so eleven measures are reported where [2] reports twelve. [TODO: This measure is being added. When it lands, update the count in the abstract, the contributions list, Sec. III-D, Sec. IV-D and the Fig. 2 caption — and this paragraph goes away entirely, since the percent-timing variants are omitted by [2] too and need no explanation from us.]

E. Latency Model

We report processing time per trial: the wall-clock interval from the end of a trial’s recording to the emission of its measures, on stated hardware. Per trial rather than per cohort, so that the figure is comparable across studies of different size, and because it is the form of the question a clinic asks — whether a result arrives within the session that produced it. [TODO: Measurement, not definition: per-stage wall-clock and the total, the hardware, and whether recording and inference share a GPU. Reported in Sec. IV-B against the per-trial cost implied by [2], which needed roughly twelve GPU-days for 1160 trials plus thirty CPU-hours of inverse kinematics on the optical side.]

F. Validation Protocol

All measures are computed from our markerless keypoints alone; the optical system contributes only the values compared against. Where our definitions depart from [2] (Sec. III-D), the reference is computed from the optical markers using those same definitions, so that the comparison isolates pose error rather than reporting a difference of definition.

Because a marker on the skin and a detected keypoint label different points on the body, a constant offset between the two is fitted per landmark, in the body and arm frames, and applied before the angles are compared. This is a landmark-matching step and not a correction of our estimate: neither system is positionally authoritative, since the optical markers carry their own placement error, and comparing unmatched landmarks reports that mismatch as though it were pose error. The fitted offsets are not themselves reported. Within each participant the fit is validated on trials held out from it, where agreement falls within 0.007 of the in-sample value. [TODO: Generalisation ACROSS participants is untested. Until it is, this is a within-participant landmark match and must not be described as transferable; a deployed version would fit against a biomechanical model rather than optical keypoints.]

The two systems are aligned in time by cross-correlating wrist speed, and the optical series is shifted onto the video timebase by the resulting lag. For the per-frame trajectory comparison a static offset is estimated per trial and removed before agreement is computed, as in [2]. Unlike the landmark offsets above, this one is reported: a consistent offset is correctable and a variable one is not, so its trial-to-trial spread is a result in itself (Tables I and II). Per-frame agreement is then Pearson r and RMSE on the offset-removed series, alongside the offset and the lag. Per-measure agreement is Spearman r_s over trials and r_{av} over per-participant, per-arm averages, following [2]. Two alignments are therefore reported for the trajectories. Removing a static offset is the like-for-like comparison with [2], but it subtracts one constant per trial and so leaves the pose-dependent part of the landmark discrepancy inside the residual. Landmark matching removes that part as well, and is consequently the better estimate of agreement; it is the alignment used for the measures throughout.

Both correlations pool across participants and arms, following [2].

¹[TODO: repository URL]

Fig 3 — Exemplary kinematic trajectories (P07): MMC vs OMC

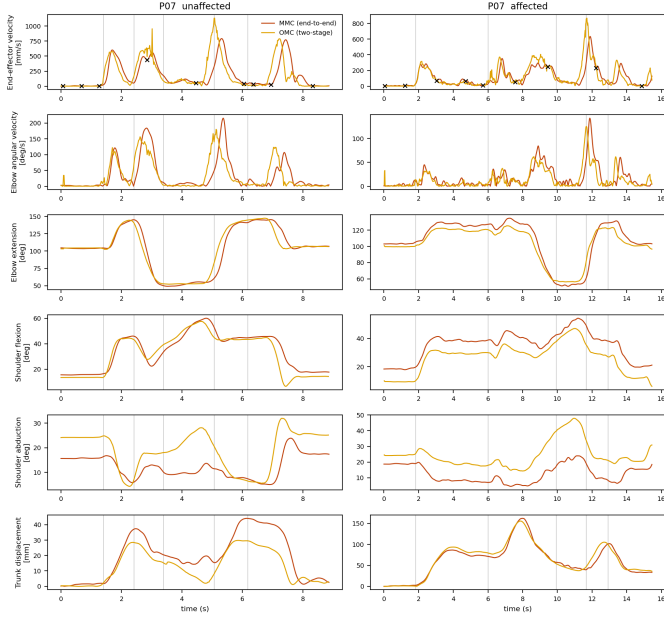


Fig. 1. Exemplary kinematic trajectories for the affected and unaffected arm, MMC versus OMC, with drinking-task phases marked and movement units indicated on the end-effector velocity. [TODO: name participant and trials]

IV. RESULTS

A. Study Population

[TODO: Table: demographics, impairment scores, trial counts, exclusions with reasons. 826/837 recorded repetitions were scorable.]

B. Latency

[TODO: Table: per-stage and total processing time per trial, with the hardware. Compare against the per-trial cost implied by [2] (twelve GPU-days for 1160 trials, plus thirty CPU-hours of inverse kinematics). NOT YET MEASURED — this table does not exist and is the main gap between the current results and the paper's claim.]

C. Kinematic Trajectory Agreement

Table I reports per-frame agreement for the six kinematic trajectories, and Table II the within-participant variability of the static bias. Fig. 1 shows exemplary trajectories for one participant. [TODO: Prose: elbow extension and shoulder flexion agree closely; the velocity trajectories sit near $r = 0.75$; shoulder abduction on the unaffected arm is the weakest. Say what the bias columns mean — a consistent offset is correctable, a variable one is not, which is what Table II addresses.]

D. Movement Quality Measures

Table III and Fig. 2 report agreement for the eleven movement-quality measures. [TODO: Prose, to be written against the REGENERATED numbers. Total movement time is phase-derived and, because both systems share optical phases

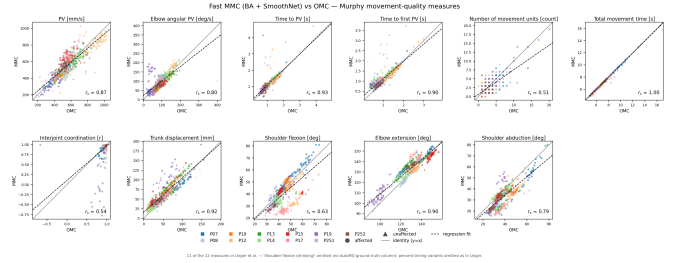


Fig. 2. Movement-quality measures, MMC versus OMC, per participant and arm. Eleven measures; see Sec. III-D for the two omitted by construction.

here, is near tautological at $r_s = 1.00$ — report it as a consistency check, not as validation of pose. Once the angles are landmark-matched only interjoint coordination and number of movement units sit low, and both are the reference's two weakest as well (Sec. V). Do NOT attribute low shoulder flexion to the occluded hip keypoint: occlusion is why the body frame is frozen, but the residual it leaves is landmark correspondence, symmetric between the systems. These are floors of the construction, not of the reconstruction. On inter-joint specifically: its reference values are near-constant across a cohort of largely intact coordination, so a rank correlation has little true variance to track and reads low even where the systems agree on every trial — report it as agreement, the fraction of trials within a tolerance plus the bias, and check the spread of the reference values before reading a low r_s as error.]

E. Phase Segmentation

The measure comparison above gives both systems the same optical phase windows, so it says nothing about segmenting from the cup. Table IV reports that separately: the same rule run on the markerless and on the optical cup, paired per trial. The quantity is the disagreement between the two inputs rather than the error against the reference phases, because a boundary bias shared by both sides cancels when markerless measures are compared against optical ones, and so is not what limits the comparison.

The segmenter declines 37 of 636 trials (5.8%). On these no cup frame survives the three-camera floor, the wrist-to-cup distance is flat, and the grasp collapses onto its floor at reach onset. Those boundaries are not measurements and are reported as a failure rate rather than scored: including them leaves the medians unchanged but puts a four-second tail on release (p_{95} 240 frames against 15), which describes the absence of a cup track rather than the accuracy of a boundary. The failure is detectable without ground truth, from the cup track alone, which is what makes reporting it possible in deployment.

On the remainder, the boundaries that bound the reaching phase agree closely — reach onset to within 33 ms and release to within 50 ms at the median — while the two drink boundaries are the weakest, at 133 ms with a fifth of trials beyond 0.25 s. [TODO: say what follows for the measures: which of the eleven are windowed by the drink boundaries,

TABLE I
MEDIAN AND [IQRs] FOR RMSE, r , BIAS, AND TIME LAG OF KINEMATIC TRAJECTORIES, CALCULATED ACROSS ALL PARTICIPANTS AND DIFFERENTIATING BETWEEN THE AFFECTED AND UNAFFECTED ARMS.

Kinematic	Value	Unaffected arm	Affected arm
End-effector velocity	r	0.75 [0.65, 0.84]	0.77 [0.71, 0.82]
	RMSE (mm/s)	141.26 [123.01, 173.38]	133.58 [116.44, 156.92]
	Bias (mm/s)	-24.98 [-30.07, -15.12]	-23.90 [-31.77, -12.13]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.15]
Elbow angular velocity	r	0.75 [0.65, 0.84]	0.74 [0.65, 0.81]
	RMSE (deg/s)	28.28 [23.31, 35.12]	26.68 [21.11, 32.93]
	Bias (deg/s)	2.02 [0.95, 3.03]	2.31 [1.28, 3.19]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.17]
Elbow extension	r	0.96 [0.91, 0.97]	0.96 [0.94, 0.98]
	RMSE (deg)	9.06 [7.09, 11.84]	9.18 [7.22, 10.95]
	Bias (deg)	0.55 [-2.12, 4.20]	-0.31 [-5.13, 3.14]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.17]
Shoulder flexion	r	0.87 [0.81, 0.93]	0.90 [0.77, 0.93]
	RMSE (deg)	7.79 [5.29, 13.37]	8.14 [5.40, 11.73]
	Bias (deg)	-2.87 [-8.82, 4.67]	-1.57 [-11.93, 3.01]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.17]
Shoulder abduction	r	0.50 [0.17, 0.75]	0.70 [0.56, 0.84]
	RMSE (deg)	4.78 [3.75, 7.39]	5.19 [3.45, 6.14]
	Bias (deg)	-8.77 [-11.96, -6.40]	-7.53 [-10.57, -5.33]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.17]
Trunk displacement	r	0.86 [0.61, 0.94]	0.93 [0.66, 0.96]
	RMSE (mm)	7.75 [5.73, 10.64]	7.32 [6.17, 10.29]
	Bias (mm)	8.93 [5.14, 14.37]	10.65 [7.21, 14.18]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.17]

TABLE II
MEAN IQR OF THE STATIC BIAS ACROSS PARTICIPANTS, I.E. THE WITHIN-PARTICIPANT TRIAL-TO-TRIAL VARIABILITY OF THE OFFSET BETWEEN MMC AND OMC.

Kinematic	Unaffected arm	Affected arm
End-effector velocity (mm/s)	7.31	10.88
Elbow angular velocity (deg/s)	2.05	1.48
Elbow extension (deg)	2.68	1.63
Shoulder flexion (deg)	3.31	2.66
Shoulder abduction (deg)	1.55	1.21
Trunk displacement (mm)	4.10	4.73

TABLE III
MOVEMENT-QUALITY MEASURES: SPEARMAN CORRELATION BETWEEN MMC AND OMC ACROSS ALL TRIALS (r_s) AND ON PER-PARTICIPANT, PER-ARM AVERAGED TRIALS (r_{av}).

Movement quality measure	r_s	r_{av}	n
PV [mm/s]	0.87	0.93	697
Elbow angular PV [deg/s]	0.80	0.76	695
Time to PV [s]	0.93	0.93	697
Time to first PV [s]	0.90	0.94	697
Number of movement units [n]	0.51	0.72	697
Total movement time [s]	1.00	1.00	697
Interjoint coordination	0.54	0.65	660
Trunk displacement [mm]	0.92	0.94	695
Shoulder flexion [deg]	0.63	0.65	660
Elbow extension [deg]	0.90	0.88	695
Shoulder abduction [deg]	0.79	0.83	660

and therefore how much of the segmentation cost in Table III is attributable to them.]

F. Ablations

[TODO: PRIMARY TABLE — the four-variant grid already run by `score_vs_automq.py`: pipeline (DLT, no smoothing), +savgol, +smoothnet, BA+smoothnet, all 11 measures on the full cohort. +smoothnet→BA+smoothnet isolates the bundle adjustment; pipeline→+smoothnet isolates the smoother; savgol shows whether the gain is SmoothNet or smoothing in general. BLOCKED: needs `out/automq/score_vs_automq.csv` copied to this machine — only the BA+smoothnet rows are committed. Known from `relax_gate_*.csv`: plain DLT → BA+SmoothNet helps 6 of 11 measures (median +0.04), all of it in the shape and derivative measures (interjoint 0.22 → 0.51, abduction 0.59 → 0.72), and nothing in

the position measures, which plain DLT already scores at 0.89–1.00.]

[TODO: Secondary, if room: bone-length prior on/off; camera-dropout and miscalibration augmentation (`augment*.csv`); segmentation factors (`seg_factors.csv`). Select rather than dump — one table of the choices a reimplementer must get right.]

V. DISCUSSION

[TODO: What the numbers support and what they do not. Once the angles are landmark-matched, only two measures sit low — interjoint coordination and number of movement units — and both are known to be fragile independently of

TABLE IV
PHASE-BOUNDARY AGREEMENT BETWEEN MARKERLESS AND OPTICAL
INPUT TO THE SAME SEGMENTER, PAIRED PER TRIAL. EXCLUDES THE
5.8% OF TRIALS ON WHICH THE SEGMENTER IS DEGENERATE FOR WANT
OF A USABLE CUP TRACK (SEE TEXT).

Boundary	n	Median (ms)	p_{90} (ms)	> 0.25 s (%)
Reach onset	599	33	87	2.7
Grasp	599	100	270	11.4
Drink onset	597	133	523	20.8
Drink offset	597	133	300	16.1
Release	599	50	167	4.8
Settle	599	50	237	9.5

markerless capture: [2] reports them as its own two weakest (0.71 and 0.77, against 0.87–0.98 for everything else) and proposes replacing movement units with log dimensionless jerk, and the study’s own processing treats its movement-unit implementation as unsettled for the same reason. So the weak pair is a property of the measures, not of the modality. Add the landmark-offset result here as evidence for the same point from the other direction: fitting a per-landmark offset lifts the angles to 0.94–0.96, so the residual was landmark correspondence rather than pose error, and the optical side is not positionally authoritative either — [2] makes the same argument by citation, that 1–2 cm of marker registration uncertainty moves a joint angle by 5–10 degrees. Then: where the remaining error lives, and the clinical implication of same-session feedback.]

VI. LIMITATIONS

[TODO: Disclose the calibration scale error: a displacement-proportional $\approx 10\%$ inflation of every 3D distance and velocity the pipeline reports, body keypoints included, which belongs to the camera calibration rather than to any tracker. Correlational measures are insensitive to it; absolute ones are not. (The cup’s own ≈ 60 mm rim-vs-centroid offset is constant and cancels in the plateau detector, so it needs no mention unless a cup number is reported.)]

[TODO: Cohort size and composition; single site; dependence on camera placement and calibration quality, including that eight of eleven units had to give up one or two cameras before they were usable; measures that could not be validated for lack of a ground-truth column; and that geometric angles are not anatomical joint angles.]

VII. CONCLUSION

[TODO: Two or three sentences.]

ACKNOWLEDGMENT

[TODO: funding, DELTA study data providers]

[TODO: REFERENCES: refs.bib holds three entries and needs the rest — YOLO (pose and segmentation), UETTrack-B, SmoothNet, the Alt Murphy planar angle definitions, prior real-time markerless work, and OpenSim/Rajagopal plus MuJoCo/Hamner to characterise what [2] does that we do not. Full list in the comment block at the end of refs.bib. Verify

every field against the publisher record; none of the current three has been checked.]

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- [3] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.